

Analisis Asesmen Kognitif Menggunakan Partial Credit Model

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```
library(eRm) # digunakan untuk pemodelan rasch
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(readxl) # digunakan untuk membaca file excel
library(psych)
```

Attaching package: 'psych'

The following object is masked from 'package:eRm':

sim.rasch

```
library(ggplot2) # digunakan untuk membuat grafik
```

Attaching package: 'ggplot2'

The following objects are masked from 'package:psych':

%+%, alpha

```
library(tidyr) # digunakan untuk merapikan data
```

```
library(flextable)
```

```
data <- read_excel("baruuu.xlsx") # membaca data menggunakan fungsi readxl dan menyimpannya di datafr
rasch_model_pcm <- PCM(data)
```

0.1 Uji Invarians

```
lrt_mean_split <- LRtest(rasch_model_pcm, splitter = "mean")
```

Warning in LRtest.Rm(rasch_model_pcm, splitter = "mean"):
The following items were excluded due to inappropriate response patterns within subgroups:
S1 S3 S5 S12 S4 S9 S2 S7 S8

Full and subgroup models are estimated without these items!

```
lrt_mean_split
```

Andersen LR-test:
LR-value: 10.051
Chi-square df: 11
p-value: 0.526

0.2 Uji Unidimensionality

```
mloef_median <- MLoef(rasch_model_pcm, splitter = "median")
mloef_median
```

Martin-Loef-Test (split criterion: median)
LR-value: 166.622
Chi-square df: 527
p-value: 1

0.3 Uji Validitas Soal

```
itemfit(person.parameter(rasch_model_pcm))
```

Itemfit Statistics:

	Chisq	df	p-value	Outfit	MSQ	Infit	MSQ	Outfit	t	Infit	t	Discrim
S1	76.388	101	0.968	0.749	0.785	-1.909	-2.282	0.606				
S2	93.314	101	0.694	0.915	0.962	-0.521	-0.204	0.249				
S3	90.663	101	0.760	0.889	0.914	-0.944	-0.724	0.432				
S4	101.708	101	0.462	0.997	0.994	0.018	-0.014	0.276				
S5	108.038	101	0.298	1.059	1.050	0.517	0.458	0.215				
S6	96.788	101	0.600	0.949	0.916	-0.258	-0.538	0.347				
S7	179.973	101	0.000	1.764	1.429	3.980	3.445	-0.041				
S8	100.127	101	0.506	0.982	0.994	-0.032	0.030	0.106				
S9	75.659	101	0.972	0.742	0.778	-1.618	-1.389	0.529				
S10	82.940	101	0.905	0.813	0.800	-1.438	-1.809	0.530				
S11	73.544	101	0.982	0.721	0.724	-2.576	-2.589	0.665				
S12	88.530	101	0.808	0.868	0.877	-0.892	-0.955	0.422				

0.4 Person Fit

```
personfit(person.parameter(rasch_model_pcm))
```

Personfit Statistics:

	Chisq	df	p-value	Outfit	MSQ	Infit	MSQ	Outfit	t	Infit	t
P1	13.442	11	0.265	1.120	1.651	0.40	1.19				
P2	6.028	11	0.871	0.502	0.465	-1.24	-1.41				
P3	6.596	11	0.831	0.550	0.735	-1.21	-0.76				
P4	16.825	11	0.113	1.402	1.084	1.04	0.36				
P5	21.336	11	0.030	1.778	1.475	1.68	1.30				
P6	6.432	11	0.843	0.536	0.704	-1.28	-0.90				
P7	34.835	11	0.000	2.903	2.491	3.30	2.73				
P8	11.848	11	0.375	0.987	0.850	0.09	-0.35				
P9	23.668	11	0.014	1.972	1.501	1.87	1.15				
P10	20.738	11	0.036	1.728	1.443	1.64	1.30				
P11	6.896	11	0.807	0.575	0.998	-0.96	0.15				
P12	7.583	11	0.750	0.632	0.671	-0.90	-0.95				
P13	8.910	11	0.630	0.743	0.828	-0.60	-0.44				
P14	7.698	11	0.740	0.641	0.907	-0.84	-0.13				
P15	12.082	11	0.357	1.007	0.934	0.15	-0.06				
P16	9.887	11	0.541	0.824	1.239	-0.24	0.61				
P17	6.771	11	0.817	0.564	0.760	-1.13	-0.63				
P18	7.413	11	0.765	0.618	0.736	-1.03	-0.74				
P19	11.214	11	0.425	0.935	0.830	-0.03	-0.40				
P20	6.069	11	0.869	0.506	0.536	-1.23	-1.15				
P21	13.437	11	0.266	1.120	1.065	0.42	0.30				
P22	5.742	11	0.890	0.478	0.470	-1.45	-1.79				
P23	10.733	11	0.466	0.894	1.343	-0.07	0.79				
P24	11.082	11	0.436	0.923	1.015	-0.07	0.15				
P25	2.302	11	0.997	0.192	0.213	-2.25	-2.05				
P26	11.341	11	0.415	0.945	1.044	-0.01	0.24				
P27	7.090	11	0.792	0.591	0.672	-1.12	-1.01				
P28	12.047	11	0.360	1.004	1.170	0.14	0.57				
P29	16.094	11	0.138	1.341	0.963	0.87	0.03				
P30	10.566	11	0.480	0.881	0.980	-0.20	0.06				
P31	16.671	11	0.118	1.389	1.673	0.75	1.23				
P32	9.634	11	0.564	0.803	0.627	-0.13	-0.61				
P33	7.455	11	0.761	0.621	0.996	-0.85	0.14				
P34	12.184	11	0.350	1.015	1.430	0.18	1.02				
P35	6.240	11	0.857	0.520	0.576	-1.22	-1.12				
P36	10.472	11	0.489	0.873	1.020	-0.18	0.19				
P37	14.323	11	0.216	1.194	0.999	0.59	0.11				
P38	10.468	11	0.489	0.872	0.595	-0.21	-1.35				
P39	5.845	11	0.883	0.487	0.345	-1.53	-2.20				
P40	8.104	11	0.704	0.675	0.627	-0.83	-1.04				
P41	5.235	11	0.919	0.436	0.445	-1.75	-1.71				
P42	8.273	11	0.689	0.689	0.691	-0.76	-0.74				
P43	14.782	11	0.193	1.232	1.177	0.69	0.56				
P44	10.420	11	0.493	0.868	1.134	-0.22	0.50				
P45	12.008	11	0.363	1.001	1.558	0.15	1.18				
P46	10.701	11	0.469	0.892	1.017	-0.16	0.16				
P47	5.727	11	0.891	0.477	0.454	-1.58	-1.83				
P48	12.695	11	0.314	1.058	1.032	0.28	0.21				
P49	10.123	11	0.519	0.844	1.045	-0.25	0.24				

P50	11.443	11	0.407	0.954	0.912	0.01	-0.15
P51	20.102	11	0.044	1.675	1.478	1.25	0.95
P52	5.670	11	0.894	0.473	0.455	-1.35	-1.45
P53	9.263	11	0.598	0.772	0.790	-0.39	-0.33
P54	8.504	11	0.668	0.709	0.546	-0.67	-1.53
P55	8.835	11	0.637	0.736	0.703	-0.48	-0.56
P56	9.366	11	0.588	0.781	0.772	-0.39	-0.44
P57	5.253	11	0.918	0.438	0.655	-1.68	-1.10
P58	5.038	11	0.929	0.420	0.440	-1.84	-1.81
P59	5.863	11	0.882	0.489	0.752	-1.38	-0.60
P60	10.035	11	0.527	0.836	1.169	-0.26	0.53
P61	15.659	11	0.154	1.305	1.555	0.84	1.32
P62	20.201	11	0.043	1.683	1.329	1.58	1.02
P63	8.203	11	0.695	0.684	0.603	-0.78	-1.03
P64	10.118	11	0.520	0.843	0.868	-0.28	-0.32
P65	6.799	11	0.815	0.567	0.742	-1.06	-0.57
P66	8.831	11	0.637	0.736	0.954	-0.61	-0.03
P67	11.272	11	0.421	0.939	1.086	-0.02	0.34
P68	17.588	11	0.092	1.466	1.368	0.95	0.79
P69	10.345	11	0.500	0.862	0.959	-0.21	0.01
P70	16.397	11	0.127	1.366	1.146	0.87	0.45
P71	16.864	11	0.112	1.405	1.212	1.04	0.72
P72	16.356	11	0.128	1.363	1.366	0.97	0.98
P73	10.233	11	0.510	0.853	0.978	-0.20	0.09
P74	35.928	11	0.000	2.994	2.340	2.40	2.36
P75	6.612	11	0.830	0.551	0.509	-0.70	-1.21
P76	13.174	11	0.282	1.098	0.944	0.37	0.00
P77	10.846	11	0.456	0.904	0.941	-0.13	-0.06
P78	10.985	11	0.445	0.915	1.336	-0.04	0.81
P79	9.241	11	0.600	0.770	0.861	-0.44	-0.23
P80	7.859	11	0.726	0.655	0.635	-0.89	-0.97
P81	21.123	11	0.032	1.760	1.751	1.62	1.60
P82	7.754	11	0.735	0.646	0.754	-0.87	-0.69
P83	12.554	11	0.323	1.046	1.179	0.25	0.62
P84	19.129	11	0.059	1.594	1.642	1.34	1.42
P85	6.343	11	0.850	0.529	0.453	-1.06	-1.46
P86	14.721	11	0.196	1.227	1.034	0.68	0.21
P87	35.898	11	0.000	2.992	2.434	3.35	3.07
P88	15.674	11	0.154	1.306	1.253	0.80	0.70
P89	9.713	11	0.556	0.809	0.914	-0.36	-0.14
P90	9.806	11	0.548	0.817	0.668	-0.17	-0.50
P91	7.098	11	0.791	0.592	0.489	-0.98	-1.45
P92	11.341	11	0.415	0.945	0.893	-0.02	-0.17
P93	8.384	11	0.679	0.699	0.588	-0.70	-1.05
P94	4.518	11	0.952	0.376	0.326	-1.51	-1.61
P95	6.161	11	0.862	0.513	0.500	-1.39	-1.42
P96	10.010	11	0.529	0.834	1.066	-0.32	0.30
P97	19.421	11	0.054	1.618	1.824	1.43	1.76
P98	12.333	11	0.339	1.028	1.097	0.20	0.38
P99	5.681	11	0.894	0.473	0.463	-1.50	-1.92
P100	8.405	11	0.677	0.700	0.685	-0.66	-0.71
P101	8.307	11	0.686	0.692	0.746	-0.74	-0.74
P102	12.350	11	0.338	1.029	0.960	0.20	0.03

0.5 Tingkat Kesukaran Soal

```
param_threshold <- thresholds(rasch_model_pcm)
print(param_threshold)
```

Design Matrix Block 1:

	Location	Threshold 1	Threshold 2	Threshold 3	Threshold 4
S1	1.13996	2.14635	-0.61988	-1.13939	4.17276
S2	-0.26814	-0.72817	-1.61408	-0.21779	1.48748
S3	1.70287	0.44718	2.95857	NA	NA
S4	-0.08668	-2.91431	0.30488	0.52865	1.73406
S5	0.64538	0.04046	0.18169	0.74833	1.61105
S6	-0.02914	-0.54046	-0.52178	-0.98036	1.92603
S7	0.29946	3.84541	-2.83638	1.30301	-1.11422
S8	0.21198	1.26632	-3.50799	2.05951	1.03006
S9	0.14619	1.08858	-3.16922	0.56570	2.09971
S10	-0.03455	0.05970	-0.10812	-0.23645	0.14665
S11	0.20022	-0.42339	-0.18102	0.49899	0.90629
S12	0.19040	-1.16204	0.26086	-1.05675	2.71951

0.6 Uji Reliabilitas Separasi

```
# 1. PERSON RELIABILITY
person_pars <- person.parameter(rasch_model_pcm)
p_rel_obj <- SepRel(person_pars)
person_reliability <- p_rel_obj$sep.rel

# 2. ITEM RELIABILITY
# eRm stores item parameters in $betapar as easiness values.
# Multiplying by -1 turns them into difficulties.
item_difficulties <- -1 * rasch_model_pcm$betapar

# Extract the standard errors of those item parameters
item_se <- rasch_model_pcm$se.beta

# Compute item variance and mean error variance
observed_item_var <- var(item_difficulties)
mean_item_error_var <- mean(item_se^2)

# Rasch Item Reliability Formula: (Observed Var - Error Var) / Observed Var
item_reliability <- (observed_item_var - mean_item_error_var) / observed_item_var

# 3. PRINT RESULTS
cat("Rasch Reliability Metrics:\n")
```

Rasch Reliability Metrics:

```
cat("Person Reliability (Separation): ", round(person_reliability, 3), "\n")
```

Person Reliability (Separation): 0.62

```
cat("Item Reliability (Separation) : ", round(item_reliability, 3), "\n")
```

Item Reliability (Separation) : 0.905

0.7 Rangkuman Hasil Uji

```

thresh_obj <- thresholds(rasch_model_pcm)

df_params <- data.frame(
  Item = rownames(thresh_obj$threshtable[[1]]),
  Calibration = round(
    thresh_obj$threshtable[[1]][, "Location"],
    6
  ),
  stringsAsFactors = FALSE
)

fit_obj <- itemfit(person.parameter(rasch_model_pcm))

df_fit <- data.frame(
  Item = rownames(thresh_obj$threshtable[[1]]),
  Infit_MSQ = as.numeric(fit_obj$i.infitMSQ),
  Outfit_MSQ = as.numeric(fit_obj$i.outfitMSQ),
  Infit_ZSTD = as.numeric(fit_obj$i.infitZ),
  Outfit_ZSTD = as.numeric(fit_obj$i.outfitZ),
  PTMEA = as.numeric(fit_obj$i.disc),
  Discrim = as.numeric(fit_obj$i.disc)
)

final_table <- data.frame(
  Item = df_params$Item,
  Calibration = round(df_params$Calibration, 4),
  Infit_MSQ = round(df_fit$Infit_MSQ[match(df_params$Item, df_fit$Item)], 4),
  Outfit_MSQ = round(df_fit$Outfit_MSQ[match(df_params$Item, df_fit$Item)], 4),
  Infit_ZSTD = round(df_fit$Infit_ZSTD[match(df_params$Item, df_fit$Item)], 4),
  Outfit_ZSTD = round(df_fit$Outfit_ZSTD[match(df_params$Item, df_fit$Item)], 4),
  PTMEA = round(df_fit$PTMEA[match(df_params$Item, df_fit$Item)], 4),
  Discrim = round(df_fit$PTMEA[match(df_params$Item, df_fit$Item)], 4)
)

print(final_table)

```

	Item	Calibration	Infit_MSQ	Outfit_MSQ	Infit_ZSTD	Outfit_ZSTD	PTMEA	Discrim
1	S1	1.1400	0.7849	0.7489	-2.2819	-1.9091	0.6064	0.6064
2	S2	-0.2681	0.9616	0.9148	-0.2035	-0.5206	0.2493	0.2493
3	S3	1.7029	0.9138	0.8888	-0.7237	-0.9437	0.4325	0.4325
4	S4	-0.0867	0.9936	0.9971	-0.0143	0.0181	0.2758	0.2758
5	S5	0.6454	1.0500	1.0592	0.4577	0.5171	0.2155	0.2155
6	S6	-0.0291	0.9156	0.9489	-0.5383	-0.2576	0.3469	0.3469
7	S7	0.2995	1.4291	1.7644	3.4452	3.9805	-0.0410	-0.0410
8	S8	0.2120	0.9942	0.9816	0.0298	-0.0323	0.1056	0.1056
9	S9	0.1462	0.7775	0.7418	-1.3892	-1.6176	0.5295	0.5295
10	S10	-0.0346	0.8002	0.8131	-1.8085	-1.4383	0.5304	0.5304
11	S11	0.2002	0.7244	0.7210	-2.5890	-2.5761	0.6653	0.6653
12	S12	0.1904	0.8775	0.8679	-0.9553	-0.8921	0.4219	0.4219